Multiphase Trajectory Optimization of Launch Vehicles with Booster Recovery

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Launch vehicle design is a multidisciplinary process in which propulsion, structural mass, aerodynamics, and trajectory decisions interact to determine overall performance. Booster recovery introduces additional coupling between ascent performance and the propellant and flight conditions required for return and landing. This paper develops a multiphase trajectory optimization framework that maximizes the payload mass delivered to a prescribed orbit by a two-stage launch vehicle with a recoverable first stage. The formulation jointly optimizes atmospheric ascent, upper-stage orbital injection, and booster return, linking the flight phases through staging conditions and propellant allocation. The framework incorporates the dominant physical models and flight constraints for expendable, downrange recovery, and return-to-launch-site missions. Predicted trajectories show good agreement with publicly available Falcon 9 telemetry. Sensitivity studies quantify how vehicle design parameters affect payload capability under different recovery strategies.

I. Introduction

Launch vehicle design is a multidisciplinary process involving propulsion, structures, aerodynamics, guidance and control, thermal protection, and other interacting subsystems [1]. These interactions require an iterative design process in which subsystem choices are repeatedly evaluated against overall vehicle performance [2]. For each candidate configuration, engineers must determine the payload mass that can be delivered to a specified orbit while satisfying physical and operational constraints. Evaluating a configuration using a prescribed guidance law can restrict the trajectories considered and obscure whether a performance limitation arises from the vehicle design or the chosen guidance parameterization. Direct trajectory optimization provides a systematic means of selecting control inputs and event times subject to vehicle dynamics and constraints [3]. Incorporating this analysis into the design process allows candidate configurations to be evaluated using optimized trajectories. Repeating the optimization as design parameters vary then reveals performance sensitivities that can guide subsequent design iterations. Booster recovery introduces additional coupling between vehicle design and trajectory performance. Propellant retained for recovery is unavailable for ascent, while the position, velocity, and mass at stage separation determine both upper-stage initial conditions and booster return requirements. Consequently, changes to a vehicle parameter can affect orbital delivery and recovery

simultaneously. Assessing these effects requires the ascent, orbital injection, and recovery trajectories to be considered within a common formulation.

Trajectory optimization for reusable launch vehicles has been investigated through direct nonlinear programming, pseudo-spectral methods, and convexification approaches. Ma et al. [4] developed a direct trajectory optimization framework for a two-stage reusable rocket that simultaneously addresses payload delivery and vehicle recovery. Garrido and Sagliano [5] formulated multistage ascent and descent problems using a multiphase optimal control framework with Radau pseudospectral transcription. Gettatelli et al. [6] combined lossless and successive convexification to optimize ascent through orbital injection together with first-stage landing. These studies demonstrate the importance of representing ascent and recovery within a coupled trajectory formulation. Related research has incorporated trajectory analysis into multidisciplinary vehicle design. Briese et al. [7] developed a modular modeling framework that integrates propulsion, aerodynamics, environmental effects, and vehicle dynamics with combined ascent and descent optimization. Dresia et al. [8] combined subsystem mass estimates, propulsion models, and ascent and landing propellant calculations to investigate how propellant selection and optimization objectives influence reusable launcher design. Jo and Ahn [9] integrated vehicle sizing and ascent/descent trajectory optimization with a life-cycle cost model.

This paper develops a multiphase optimal control formulation that maximizes the payload mass delivered to a prescribed orbit by a two-stage launch vehicle. Atmospheric ascent, upper-stage orbital injection, and booster return are coupled through staging conditions, propellant allocation, and recovery constraints. The formulation accommodates expendable flight, downrange booster landing, and return to the launch site, subject to propulsion, aerodynamic loading, acceleration, heating, and terminal constraints.

Falcon 9 is used as a case study. Optimized trajectories show good agreement with publicly available telemetry across missions with different payload masses and orbital targets. The comparisons use available altitude and speed histories and cataloged payload orbital elements, with vehicle parameters and mission-specific inputs documented for each case. The framework is subsequently used to quantify how vehicle design parameters affect payload capability under different recovery strategies, supporting preliminary design decisions. The remainder of the paper presents the physical models, multiphase optimal control formulation, and numerical solution method, followed by the telemetry comparisons and design sensitivity studies.

II. Problem Formulation

Consider a two-stage launch vehicle tasked with delivering a payload from a launch site specified by its latitude, longitude, and altitude to a target orbit specified by its perigee altitude, apogee altitude, and inclination. The remaining orbital elements are left unconstrained; launch timing may be selected to satisfy additional orbital-plane alignment requirements. For a given vehicle configuration and booster operating mode, the objective is to maximize the delivered payload mass by jointly optimizing ascent, upper-stage flight, and, where applicable, booster recovery. Payload mass

excludes upper-stage inert mass and residual propellant.

Both stages use liquid-propellant rocket engines with prescribed minimum and maximum throttle settings during powered flight. First-stage aerodynamic maneuvering is permitted within a prescribed maximum absolute angle of attack. The upper-stage thrust direction is freely selectable in three-dimensional space during powered flight.

The following environmental assumptions are adopted:

- Earth is modeled as a sphere with a central gravitational field.
- Atmospheric properties follow a standard atmosphere model.
- Winds are neglected.

Three booster operating modes are considered:

- **Expendable:** The booster is not recovered, and no recovery terminal conditions are imposed.
- **ASDS:** The booster lands on an autonomous spaceport drone ship downrange of the launch site.
- **RTLS:** The booster performs a return maneuver and lands at the launch site.

The mission comprises atmospheric ascent, upper-stage flight to the target orbit, and booster return for recovery missions. Atmospheric ascent is governed by propulsion, gravity, and aerodynamic forces. Upper-stage flight is modeled outside the atmosphere and includes two engine ignitions. Following stage separation, the upper stage undergoes a ballistic coast of prescribed duration before its first ignition, which establishes a parking orbit. The second ignition occurs at the apogee of the parking orbit and is represented by an instantaneous velocity impulse that achieves the prescribed target orbit. This assumes that the second burn of the second engine is relatively short and is a very small fraction of the orbit time. Booster recovery includes a boostback maneuver for RTLS missions, ballistic flight, atmospheric reentry with a reentry burn where required, and powered landing, subject to prescribed upper bounds on dynamic pressure and aerodynamic heat flux.

The position and velocity at stage separation initialize both upper-stage flight and booster return, while the remaining booster propellant determines the resources available for recovery. These shared separation conditions couple ascent performance to recovery feasibility and are optimized jointly. The physical models, phase dynamics, and associated constraints are defined in the following sections.

III. Physical Models

The trajectory formulation uses reduced-order physical models to capture the dominant effects on flight dynamics and performance. The vehicle's inertial acceleration results from thrust, gravity, and aerodynamic forces, with control inputs governing engine throttle and vehicle orientation. Its mass decreases as propellant is consumed. Atmospheric density and pressure influence aerodynamic loading, engine performance, and convective heating. The relative importance of these effects changes throughout flight: aerodynamic forces and heating constrain atmospheric ascent and booster reentry, while thrust and gravity dominate upper-stage flight outside the atmosphere. The coordinate systems and

individual models are defined below.

A. Coordinate Systems

Two Coordinate Systems (CS) are used to describe the flight dynamics. First-stage ascent and booster recovery are modeled in a local level azimuth-aligned (LLAA) frame with its origin at the launch site. The local x-axis lies in the tangent plane and points along the launch azimuth, while the local z-axis points radially away from earth center at the launch site. The axes remain fixed relative to Earth, and motion is restricted to the local x–z plane (it is not an inertial frame). The launch azimuth is an optimization variable.

Upper-stage flight is modeled in a three-dimensional Earth-centered inertial (ECI) frame. Its origin is at Earth's center, its Z-axis points toward the North Pole, and its X-axis lies in the equatorial plane and points toward the Greenwich meridian at liftoff. The Y-axis completes a right-handed frame. The ECI axes remain fixed in inertial space.

Given a launch site and time from launch, the transition of position and velocity between both CS is defined.

Figure 1 illustrates the relationship between the two frames. Transformations of position and velocity between the frames, including the contribution of Earth rotation to inertial velocity, are presented in Appendix **??**.

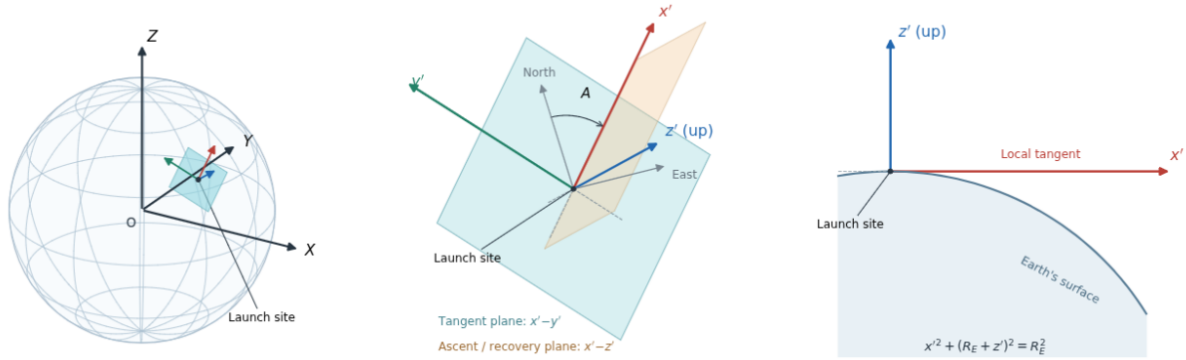


Fig. 1 Coordinate systems used in the trajectory formulation. Left: Earth-centered inertial (ECI) frame and launch-site location. Center: local level azimuth-aligned (LLAA) frame, showing the launch azimuth (A) measured from North toward East to the (x')-axis, the tangent plane (cyan), and the ascent and recovery plane (orange). Right: LLAA side view showing Earth's surface curving below the tangent (x')-axis, with (z') directed upward at the launch site.

B. Rocket Engine - Thrust

Engine thrust depends on propellant mass flow and ambient pressure. For a fixed nozzle, the thrust is expressed as [10]

$$T = \dot{m}V_e + (P_e - P_a)A_e, \tag{1}$$

where $\dot{m}$ is the positive propellant mass flow rate, V_e is the nozzle exit velocity, P_e is the nozzle exit pressure, P_a is the ambient pressure, and A_e is the nozzle exit area. Throttling changes the mass flow rate and chamber pressure, while

changes in altitude affect the ambient-pressure contribution to thrust. For first-stage ascent, a reduced-order model approximates thrust using the full-throttle vacuum and sea-level values:

$$T(f_T, P_a) = f_T \left[T_{\text{Vac}} - (T_{\text{Vac}} - T_{\text{SL}}) \frac{P_a}{P_{\text{SL}}} \right], \tag{2}$$

where P_{SL} is the reference sea-level pressure and f_T is the throttle control, bounded by

$$f_T \in [f_T^{\min}, 1]. \tag{3}$$

This expression interpolates between sea-level and vacuum thrust and assumes that thrust at a given ambient pressure varies linearly with throttle. It is an engineering approximation rather than an exact consequence of fixed-nozzle isentropic flow. The corresponding propellant mass flow model is

$$\dot{m} = f_T \frac{T_{\text{Vac}}}{I_{sp,\text{Vac}} g_0}, \tag{4}$$

where $I_{sp,\text{Vac}}$ is the full-throttle vacuum specific impulse and g_0 is standard gravitational acceleration. The vehicle mass therefore evolves according to

$$\frac{dm}{dt} = -\dot{m}. \tag{5}$$

During upper-stage powered flight, ambient-pressure effects are neglected. The thrust magnitude determines the propellant consumption through

$$\dot{m} = \frac{\|\mathbf{T}\|}{g_0 I_{sp}(f_T)}, \qquad f_T = \frac{\|\mathbf{T}\|}{T_{\text{Vac}}}, \tag{6}$$

with a linear correction for reduced specific impulse at partial throttle:

$$I_{sp}(f_T) = I_{sp,\text{Vac}} \left[\eta_{\min} + (1 - \eta_{\min}) \frac{f_T - f_T^{\min}}{1 - f_T^{\min}} \right]. \tag{7}$$

Here, $\eta_{\min}$ is the ratio of specific impulse at minimum throttle to its full-throttle value. Booster boostback uses constant vacuum specific impulse, while reentry and landing burns use constant sea-level specific impulse. These propulsion models provide thrust and propellant consumption for the phase dynamics. Their accuracy is limited by the assumed pressure and throttle dependence; nozzle flow separation and detailed combustion and turbomachinery behavior are not explicitly modeled.

C. Aerodynamic Model - Lift and Drag

During the boost phase of the launch vehicle (LV), aerodynamic forces act on its structure. The two primary aerodynamic forces are lift and drag. Lift is the component of the aerodynamic force acting perpendicular to the vehicle's velocity vector relative to the surrounding air, while drag is the component acting in the opposite direction to the velocity vector. A simple model for the lift force is a linear model:

$$F_L = \frac{1}{2}\rho V^2 \cdot S_{ref} \cdot C_{L_\alpha} \cdot \alpha = Q_{dyn} \cdot S_{ref} \cdot C_{L\alpha} \cdot \alpha \tag{8}$$

where ρ is the density of the surrounding air, V is the velocity relative to air, S_{ref} is the reference area (usually taken as the LV cross-section area), $C_{L\alpha}$ is the lift line slope of the LV and α is the angle if attack - the angle between the velocity vector and the LV longitudinal body direction. In this paper α is used as a control input during the boost phase. This will allow to control the LV trajectory shaping during the boost phase. The linear lift model is a good approximation for low angles of attack. The drag force model is consists from a shape drag at zero angle of attack and add induced drag:

$$F_D = Q_{dyn} \cdot S_{ref} \cdot (C_{D_0} + C_{D_{\alpha^2}}\alpha^2) \tag{9}$$

where C_{D_0} describing the shape drag of the vehicle and $C_{D_{\alpha^2}}$ is the induced drag coefficient. In addition, The aerodynamic model can provide an estimate to the aerodynamic loads associated with aerodynamic maneuver. When a aerodynamic body develops lift forces, it usually accompanied with significant structural bending moments which has to be limited (specially in a long body like LV). The bending moment for small angle of attack are proportional to:

$$M_{bending} \propto Q_{dyn} \cdot \alpha \tag{10}$$

D. Gravity Model

The gravity model used is the standard model of a uniform body. When expressed in ECI CS, the gravitational acceleration as a function of position vector $\vec{r}$ can be expressed as:

$$\vec{g}(r) = -\frac{\mu_e M_e}{r^3}\vec{r} = -g_0 \cdot \frac{R_e^2}{r^3} \cdot \vec{r} \tag{11}$$

where μ is the gravitational constant, M_e and R_e is the mass and radius of earth, respectively and g_0 is the gravitational acceleration at earth surface. When expressed in LLAA CS, the acceleration vector is as function of the position $[x, z]$

is:

$$g_x = -g_0 \frac{R_e^2}{r^2} \cdot sin(\theta)$$
$$g_z = -g_0 \frac{R_e^2}{r^2} \cdot cos(\theta) \tag{12}$$

where $r = \sqrt{x^2 + (R_E + z)^2}$ and $\theta = sin^{-1}(\frac{x}{R_e+z})$.

E. heating model

Aerodynamic heating is a primary driver of reentry loads, especially for a booster during the return phase. Boosters are typically oriented engine-first on entry because the nozzle/engine section is comparatively heat-tolerant; additional local thermal protection can be added to shield more sensitive structures. Even so, there is always a flight condition where the heat flux becomes excessive and risks structural damage, so operations enforce a heat-flux limit. In this work we model stagnation-point convective heating using the classical Sutton–Graves correlation [11]:

$$\dot{q}_c = \kappa_e \cdot \sqrt{\rho} v^3 \tag{13}$$

where, κ_e is an empirical coefficient that depends on booster structure properties, ρ is the air density and v is the velocity. The heat flux has units of $\frac{Watt}{m^2}$. The allowed values for the maximal allowed heat flux can be approximated from theoretical calculations or computed from test flights - like telemetry from Falcon 9 flights. The max heating flux will constraint the dynamic condition during reentry. This can be managed by preforming engines burns to slow down the booster to limit the heat flux.

IV. Equations Of Motions

The equation of motion that describe the launch of a typical launch vehicle can be divided into 2 phases. Since this paper talks about the recovery of the booster, an additional phase is added to model the reentry phase of the booster.

A. First Stage Ascent

The first stage lifts off vertically from rest relative to Earth's surface and subsequently performs a pitch-over maneuver along a launch azimuth selected by the optimization. Motion during this phase is restricted to the corresponding LLAA flight plane. Centrifugal and Coriolis accelerations associated with Earth's rotation are neglected during first-stage ascent. The state vector is defined as

$$X_1 = [x, z, v_x, v_z, m]^T \tag{14}$$

where, x, z are the horizontal and vertical position in the LLAA CS, respectively. v_x, v_z are the horizontal and vertical velocity in the LLAA CS respectively, and m is the mass of the LV. The inputs of the LV during this phase are:

$$\begin{gathered} U_1 = [\alpha, f_T]^T \\ \alpha \in [-\alpha_{min}, \alpha_{max}], \quad f_T \in [f_T^{min}, 1] \end{gathered} \tag{15}$$

where, α is the angle of attack of the LV, and f_T is the throttle of the engines. The dynamics of this phase is given as:

$$\begin{cases} \dot{x} = v_x \\ \dot{z} = v_z \\ \dot{v}_x = \dfrac{1}{m}[-\text{Drag } \cos(\gamma) - \text{Lift } \sin(\gamma) + \text{T } \cos(\gamma + \alpha)] + g \ \sin(\theta_g) \\ \dot{v}_z = \dfrac{1}{m}[-\text{Drag } \sin(\gamma) + \text{Lift } \cos(\gamma) + \text{T } \sin(\gamma + \alpha)] + g \ \cos(\theta_g) \\ \dot{m} = -\dfrac{\text{THR}^1_{\text{Vac}}}{\text{ISP}^1_{\text{Vac}}\, g_0}\, f_T \end{cases} \tag{16}$$

where,

$$\begin{aligned} \gamma &= \text{atan2}(v_z,\ v_x) \\ \text{Drag} &= Q_{\text{dyn}}\, S_{\text{ref}} \left(C_{D_0} + C_{D\alpha^2}\, \alpha^2 \right) \\ \text{Lift} &= Q_{\text{dyn}}\, S_{\text{ref}}\, C_{L\alpha}\, \alpha \\ g &= g_0\, \frac{R_0^2}{x^2 + (R_0 + z)^2} \\ \theta_g &= \arcsin\left(\frac{x}{R_0 + z} \right) \end{aligned} \tag{17}$$

The dynamics above are valid only if the velocity of the LV is above some threshold (due to the definition of γ). For this reason, the initial condition of the optimization are defined after a section of an open loop vertical ascent for a few second. This logic is a popular operational technique that helps to clear the launch site infrastructure (launch tower).

B. Second Stage

The second stage phase begins after the physical separation between the first and second stage. This phase starts at high altitudes where dynamic pressure is negligible. During this phase the LV need to reach the orbital velocity (size and direction) that matches the desired orbit. this means that the second stage needs to maneuver in a 3D space. This phase is modeled in the ECI CS. The states of this phase are:

$$X_2 = [x, y, z, v_x, v_y, v_z, m]^T \tag{18}$$

where x, y, z are the position in ECI CS, v_x, v_y, v_z are the velocity in ECI and m is the mass. The inputs for the system during this phase are:

$$U_2 = [f_x, f_y, f_z]^T \tag{19}$$

where f_x, f_y, f_z are the forces produces by the engine in ECI frame. This means that the second stage can orientate itself in any angular state to point the engine's nozzle in the appropriate direction that produce the wanted inertial force. This is usually done by a actuators that control the second stage orientation (for example, gimbled nozzle). Since the stage engine can be throttled, the maximal force allowed is given by the maximal thrust that the rocket engine can provide:

$$\sqrt{f_x^2 + f_y^2 + f_z^2} \leq \mathrm{THR}_{\mathrm{Vac}}^{(2)} \tag{20}$$

The dynamics of the phase is given by:

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = \dfrac{1}{m} f_x - g_x \\ \dot{v}_y = \dfrac{1}{m} f_y - g_y \\ \dot{v}_z = \dfrac{1}{m} f_z - g_z \\ \dot{m} = -\dfrac{\sqrt{f_x^2 + f_y^2 + f_z^2}}{\mathrm{ISP}_{\mathrm{Vac}}^2 \, g_0} \end{cases} \tag{21}$$

where,

$$\begin{aligned} g_x &= g_0 \frac{R_0^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cdot x \\ g_y &= g_0 \frac{R_0^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cdot y \\ g_z &= g_0 \frac{R_0^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cdot z \end{aligned} \tag{22}$$

The upper-stage phase is initialized from the first-stage separation state, accounting for the removal of booster mass and the prescribed ballistic coast before engine ignition. Position and velocity are transformed from the LLAA frame to the ECI frame, including the contribution of Earth's rotation to inertial velocity, as described in Appendix A.

C. Booster Return

Following stage separation, the booster follows either a RTLS trajectory or a downrange recovery trajectory (ASDS). RTLS includes a boostback burn that redirects the booster toward the launch site, whereas downrange recovery omits

this maneuver. Both profiles subsequently include ballistic flight, a reentry burn with optimized duration, and powered landing. Booster return is modeled in the same Earth-fixed LLAA frame as first-stage ascent. Coriolis and centrifugal accelerations associated with Earth's rotation are neglected. The model estimates recovery propellant requirements and their effect on payload capability; detailed attitude dynamics and small terminal guidance corrections are neglected. For ASDS recovery, the drone ship is assumed to be positioned at the landing location determined by the optimization. The state vector has the same components as the first-stage ascent state in (14). Position and velocity are continuous across separation, and separation impulses are neglected. The initial booster mass is

$$m_{\mathrm{B},0} = m_{1,f} - m_{\mathrm{upper,sep}}, \tag{23}$$

where $m_{1,f}$ is the total vehicle mass immediately before separation and $m_{\mathrm{upper,sep}}$ includes the upper-stage inert mass, its propellant, the payload, and the attached fairing. The initial booster mass therefore consists of its inert mass and retained recovery propellant. For RTLS missions, the boostback burn occurs at high altitude, where aerodynamic forces are neglected. Its control input is

$$U_{\mathrm{boostback}} = [f_x, f_z]^T, \tag{24}$$

where f_x and f_z are the thrust components along the LLAA axes. A single thrust vector is optimized and held constant throughout the burn, subject to the prescribed thrust-magnitude bound. Both thrust direction and magnitude are therefore constant in the LLAA frame, while the burn duration is an optimization variable. This approximation reduces the number of decision variables by replacing continuously varying thrust-direction and throttle histories with a constant thrust vector.

During atmospheric return, aerodynamic lift is neglected and drag is modeled using a constant recovery drag coefficient. The booster is assumed to maintain an engine-first orientation, with thrust directed opposite to its air-relative velocity. Consequently, thrust magnitude is the only control input during reentry and landing burns. Independent aerodynamic steering is not modeled. During ballistic segments, thrust and propellant consumption are zero.

Recovery trajectories are subject to dynamic-pressure and heat-flux limits. Terminal constraints enforce landing at the launch site for RTLS or at sea level for downrange recovery, together with an upper bound on touchdown speed.

V. Flight Logic and Constraints

The mission is represented by a sequence of powered and ballistic phases linked through continuity of position and velocity and the appropriate changes in mass at separation events. First-stage ascent terminates at stage separation, after which upper-stage flight and booster recovery proceed independently from their shared separation conditions. Control inputs and phase durations are optimized subject to the following operational and flight constraints.

A. First-Stage Ascent

The vehicle initially performs a prescribed vertical ascent at full throttle for 5 s. The resulting state initializes the optimized ascent trajectory. During this phase, the following constraints apply:

- Engine throttle and angle of attack satisfy their prescribed bounds:

$$f_T^{\min} \le f_T \le 1, \qquad |\alpha| \le \alpha_{\max}. \tag{25}$$

- Dynamic pressure is limited to reduce aerodynamic loading:

$$Q_{\mathrm{dyn}} \le Q_{\max}^{(1)}. \tag{26}$$

- The aerodynamic bending-load proxy satisfies

$$Q_{\mathrm{dyn}}|\alpha| \le Q_{\alpha,\max}, \tag{27}$$

  where $Q_{\alpha,\max}$ is the prescribed limit on the product of dynamic pressure and absolute angle of attack.

- The magnitude of acceleration due to nongravitational forces is bounded to limit vehicle and payload loading:

$$a_{\mathrm{ng}} = \frac{\|\mathbf{T} + \mathbf{F}_{\mathrm{aero}}\|}{m} \le a_{\max}, \tag{28}$$

  where $\mathbf{F}_{\mathrm{aero}}$ is the resultant aerodynamic force.

- Stage separation is permitted only when dynamic pressure satisfies

$$Q_{\mathrm{dyn}}(t_{\mathrm{sep}}) \le Q_{\max}^{\mathrm{sep}}. \tag{29}$$

For expendable missions, the booster mass at separation cannot fall below its inert mass. For recovery missions, the retained propellant must instead be sufficient to satisfy the subsequent recovery constraints.

B. Upper-Stage Flight

Following stage separation, the upper stage undergoes a ballistic coast of prescribed duration before its first engine ignition. The powered trajectory is divided into segments before and after fairing separation. These segments represent a single burn, with an instantaneous reduction in mass when the fairing is released. The following constraints apply:

- During powered flight, thrust magnitude satisfies

$$f_{T,2}^{\min} T_{\mathrm{Vac}}^{(2)} \le \|\mathbf{T}_2\| \le T_{\mathrm{Vac}}^{(2)}. \tag{30}$$

- Fairing separation is permitted above a prescribed minimum altitude:

$$h(t_{\text{fairing}}) \geq h_{\text{min}}^{\text{fairing}}. \tag{31}$$

- After fairing separation, payload acceleration is limited by

$$\frac{\|\mathbf{T}_2\|}{m} \leq a_{\max}. \tag{32}$$

- At the end of the first burn, the parking-orbit perigee altitude must satisfy

$$h_{p,\text{parking}} \geq h_{p,\text{min}}^{\text{parking}}. \tag{33}$$

- The parking-orbit apogee altitude must equal the target-orbit perigee altitude:

$$h_{a,\text{parking}} = h_{p,\text{target}}. \tag{34}$$

The upper stage subsequently coasts to the parking-orbit apogee, where a second burn is represented by an instantaneous velocity impulse. This maneuver establishes the target apogee and inclination while retaining the prescribed target perigee. The parking-orbit inclination is therefore not constrained to equal the target inclination before this maneuver. Sufficient propellant is reserved for the required velocity and orbital-plane changes, and the mass after the impulse must satisfy

$$m_{2,f}^{+} \geq m_{2,\text{inert}} + m_{\text{payload}}, \tag{35}$$

where $m_{2,f}^{+}$ is the upper-stage mass immediately after the impulse.

C. Booster Recovery

RTLS recovery includes a boostback burn, followed by ballistic flight, a reentry burn, a further ballistic segment, and powered landing. ASDS recovery follows the same sequence without boostback. Recovery burn and coast durations are optimization variables subject to prescribed lower bounds. The reentry-burn duration may approach its minimum permitted value when little deceleration is required. The following constraints apply:

- Boostback thrust is bounded, while reentry and landing thrust satisfy phase-specific minimum and maximum values.
- The reentry burn begins above a prescribed minimum altitude:

$$h(t_{\text{reentry}}) \geq h_{\text{min}}^{\text{reentry}}. \tag{36}$$

- During the ballistic segment following the reentry burn and during powered landing, dynamic pressure and convective heat flux satisfy

$$Q_{\mathrm{dyn}} \le Q_{\max}^{\mathrm{return}}, \qquad \dot{q}_c \le \dot{q}_{c,\max}^{\mathrm{return}}. \tag{37}$$

- RTLS terminal conditions require the booster to reach the launch-site position. ASDS terminal conditions require sea-level altitude, while the downrange landing location is determined by the optimization.
- Touchdown speed is bounded by

$$\sqrt{v_{x,f}^2 + v_{z,f}^2} \le v_{\mathrm{land}}. \tag{38}$$

- Final booster mass must be at least its inert mass:

$$m_{\mathrm{B},f} \ge m_{\mathrm{B,inert}}. \tag{39}$$

VI. Numerical Formulation and Implementation

The multiphase optimal control problem is converted into a nonlinear program using a combination of direct multiple shooting and numerical propagation. The decision variables include payload mass, launch azimuth, phase durations, and the state and control variables associated with the discretized flight segments. Vehicle parameters and mission requirements remain fixed within each optimization.

A. Discretization and Objective

For each discretized segment, the state and control profiles are represented on a uniform time grid, with piecewise-constant controls. The time step is an optimization variable that determines the segment duration through $T_j = N_j \Delta t_j$, where N_j is the prescribed number of integration intervals. The dynamics are enforced through

$$X_{j,k+1} = \Phi_j(X_{j,k}, U_{j,k}, \Delta t_j), \tag{40}$$

where j identifies the flight segment and Φ_j is a fourth-order Runge–Kutta integration map. Ballistic segments and burns with constant control inputs are propagated using internal integration substeps. The boostback thrust vector (direction in LLAA and magnitude) and reentry-burn thrust magnitude are held constant within their respective burns, while landing thrust is discretized over the landing segment.

Phase-linking constraints enforce continuity of position and velocity, coordinate transformations, and the mass changes associated with stage and fairing separation. The final orbital maneuver is evaluated analytically (instantaneous velocity pulse is assumed at the parking apogee which become final orbit perigee), and its propellant requirement is included in the upper-stage terminal mass constraint. Path constraints are imposed at designated discrete evaluation

points.

The optimization objective depends on whether the payload mass is optimized or prescribed. When the payload mass is a decision variable, the objective is to maximize the delivered payload while penalizing changes in control inputs:

$$\min_{\mathbf{z}} \; -m_{\mathrm{PL}} + J_{\mathrm{reg}}(\mathbf{z}), \tag{41}$$

where $\mathbf{z}$ collects the decision variables, m_{PL} is the payload mass, and J_{reg} is a weighted quadratic control-smoothing penalty. This penalty acts on control increments rather than time derivatives. For flight-data comparisons, the payload mass is prescribed. In this case, the objective is to maximize the upper-stage residual propellant mass after the final orbital maneuver:

$$\min_{\mathbf{z}} \; -m_{\mathrm{P,res}} + J_{\mathrm{reg}}(\mathbf{z}), \tag{42}$$

where $m_{\mathrm{P,res}}$ denotes the upper-stage propellant remaining after orbital injection.

B. Implementation and Solution

The formulation is implemented in Python using CasADi [12], which provides symbolic expressions and automatic derivatives for the nonlinear program. States, controls, and time are scaled to improve numerical conditioning. Initial trajectories are generated using approximate ascent and recovery profiles, and previously computed solutions can initialize related cases. The nonlinear program is solved using IPOPT [13]. Because the problem is nonconvex, convergence does not guarantee a globally optimal solution.

A common implementation supports expendable, ASDS, and RTLS missions. Performance comparisons and sensitivity studies are performed by changing the recovery strategy or prescribed parameters and reoptimizing the trajectories.

VII. Verification and Flight-Data Assessment

To assess the numerical correctness and predictive accuracy of the proposed framework, numerical verification is done by comparisons with publicly available Falcon 9 flight data. Flight-data comparisons assess agreement in altitude, speed, flight-event times, and available orbital elements across missions with different payloads, orbital targets, and recovery strategies.

A. Comparison with Flight Data

The optimized trajectories are compared with publicly available Falcon 9 flight telemetry. Launch broadcasts provide time histories of altitude and speed for first-stage ascent, upper-stage flight, and, where available, booster recovery. The payload mass for each scenario is taken from publicly available mission information, while the prescribed final orbit is

obtained from publicly available cataloged orbital data for the deployed payload. Missions with different payload masses, target orbits, and recovery strategies are considered to assess the framework across different operating conditions.

The comparison assumes that mission-specific range-safety and operational restrictions not represented in the formulation have a limited effect on vehicle performance. For reusable missions, additional restrictions intended to preserve the booster for subsequent flights are neglected beyond the modeled constraints, including recovery dynamic-pressure and heat-flux limits. In particular, engine operation up to the prescribed maximum throttle is permitted without additional reuse-related restrictions.

For these comparisons, payload mass and target orbit are fixed, and the trajectory optimization maximizes the propellant remaining in the upper stage at payload separation, after accounting for the final orbital maneuver, together with the control-smoothing penalty described in Section VI. With payload and upper-stage inert mass fixed, maximizing residual propellant is equivalent to maximizing the upper-stage mass remaining after orbital injection. Retained propellant is available for post-separation operations, including a deorbit maneuver; the deorbit maneuver itself is not explicitly optimized.

Calibration is performed using selected reference missions. Uncertain publicly reported vehicle parameters are adjusted, and unavailable parameters are estimated within prescribed bounds, to improve agreement between telemetry and optimized trajectories. Each candidate parameter set requires a new trajectory optimization. The calibrated vehicle parameters are then retained when assessing other missions; only mission-specific inputs, such as payload mass, target orbit, launch site, and recovery strategy, are changed. This separates parameter calibration from assessment of the framework's ability to reproduce other flight scenarios.

Before comparison, a common time offset aligns the first simulated speed with the corresponding rising-speed crossing in the telemetry, using first-stage data when available. Simulated quantities are interpolated at the telemetry timestamps, and upper-stage inertial velocities are converted to Earth-fixed speeds. The comparison uses altitude, speed, acceleration, and dynamic pressure during ascent, and altitude and speed during booster return. When acceleration or dynamic pressure is unavailable, these quantities are estimated from the speed history or from altitude and speed using the atmospheric model, respectively. These derived quantities supplement the comparison but do not constitute independent measurements. Ascent samples are restricted to intervals of increasing measured speed, while available booster-return data include both powered and ballistic segments.

Agreement is quantified using a dimensionless robust error:

$$J_{\text{data}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \frac{1}{|C_p|} \sum_{c \in C_p} \frac{1}{N_{p,c}} \sum_{k=1}^{N_{p,c}} \left(\frac{\hat{y}_{p,c,k} - y_{p,c,k}}{s_c} \right)^2 \tag{43}$$

where $\mathcal{P}$ is the set of assessed phases, C_p contains the available comparison channels in phase p, and $y_{p,c,k}$ and $\hat{y}_{p,c,k}$ denote telemetry and interpolated predictions. The normalization scales are $s_h = 100\,\text{m}$, $s_V = 5\,\text{m/s}$. Averaging within

each channel and phase prevents densely sampled channels or phases from dominating the score. Lower scores indicate closer agreement; channel-wise root-mean-square errors are also reported in physical units to support interpretation.

First we take a single scenario as baseline to estimate the parameters of Falcon 9. The selected reference scenario is Starlink Group 15-27, launched from Space Launch Complex 4 East (SLC-4E) at Vandenberg Space Force Base, California. The mission carried 27 Starlink satellites, with an estimated total payload mass of approximately 15.5 t, and used ASDS recovery. Early cataloged orbital elements indicate an insertion orbit with perigee and apogee altitudes of approximately 248 km and 264 km, respectively, at an inclination of 70°.

Figure 2 compares the optimized Starlink Group 15-27 trajectory with flight telemetry. First-stage altitude and speed closely follow the measured histories. Booster-return altitude also shows close agreement, including the maximum altitude and subsequent descent. Booster speed is slightly overpredicted before the reentry burn, while the subsequent atmospheric deceleration agrees more closely with the available measurements. Gaps in telemetry coverage limit assessment of the reentry burn and final landing maneuver.

The largest discrepancy occurs in upper-stage altitude. The optimized trajectory reaches approximately 159 km, compared with approximately 151 km in the telemetry, and remains higher during the subsequent descent. Upper-stage speed follows the measured trend but is generally overpredicted, reaching a similar terminal speed earlier. These differences indicate that the optimization reproduces the main flight trends while selecting a slightly different upper-stage trajectory.

The parameter fit for Starlink Group 15-27 yields a comparison cost of approximately 120. Table 1 compares the nominal model parameters with the fitted values. The nominal values combine publicly reported vehicle characteristics and modeling assumptions; they are not all specifications published by SpaceX. The fitted parameters should be interpreted as effective values within the proposed model, since adjustments may also compensate for uncertainties in other vehicle properties and modeling approximations.

Table 1 Nominal and fitted vehicle parameters for Starlink Group 15-27. First-stage thrust values represent the combined thrust of nine engines.

Parameter	Nominal	Fitted
First-stage sea-level specific impulse [s]	282.0	265.08
First-stage vacuum specific impulse [s]	305.0	286.70
Upper-stage vacuum specific impulse [s]	348.0	349.74
First-stage sea-level thrust [kN]	7605.0	7148.70
First-stage vacuum thrust [kN]	8226.0	7732.44
Upper-stage vacuum thrust [kN]	981.0	892.71

The fit reduces first-stage thrust and specific impulse by 6%, increases upper-stage specific impulse by 0.5%, reduces upper-stage thrust by 9%, and increases the booster-return drag coefficient by 30%. The ascent drag coefficient remains unchanged. The remaining prescribed vehicle parameters and scenario settings are summarized in Table 2. These

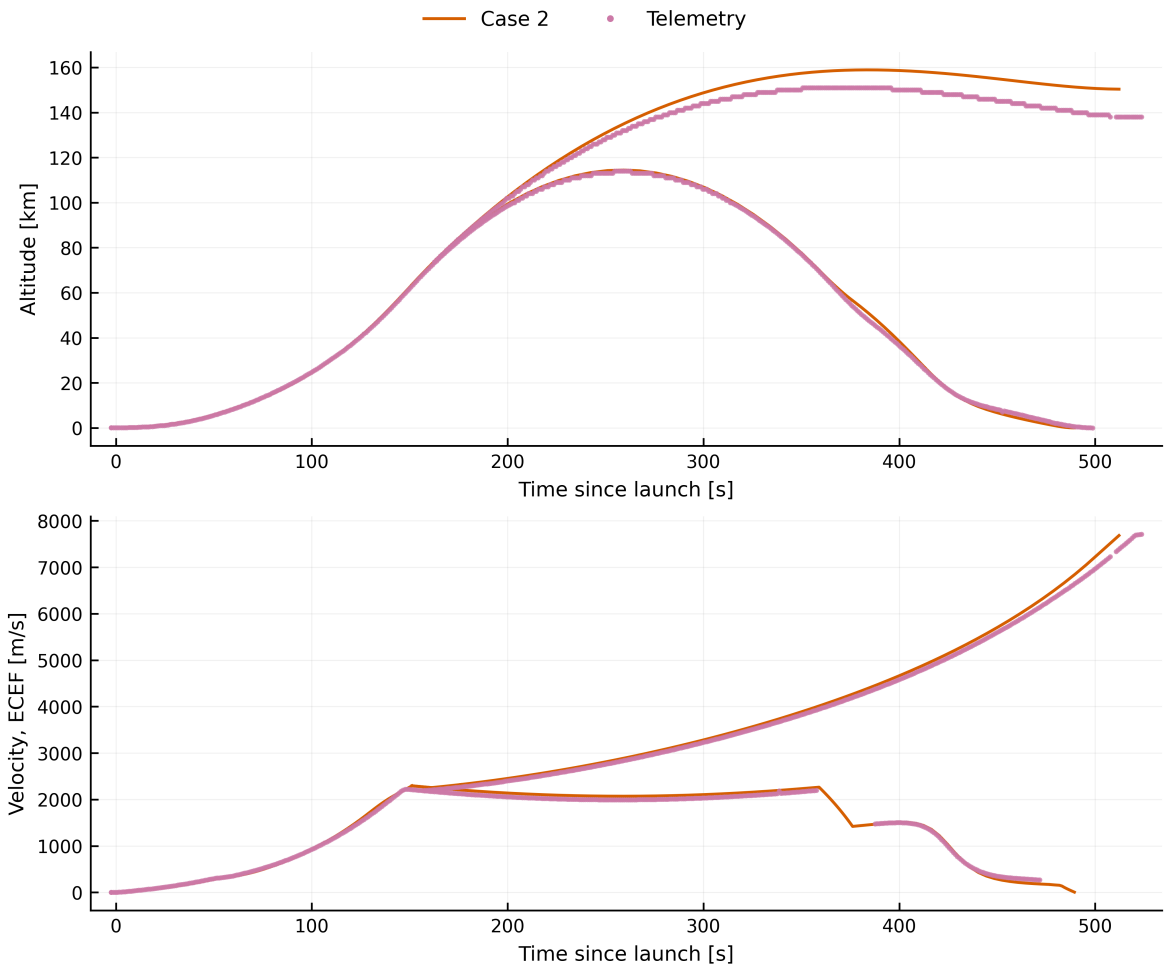


Fig. 2 Comparison of the optimized Starlink Group 15-27 trajectory (solid orange lines) with publicly available flight telemetry (magenta markers). Altitude (top) and Earth-fixed speed (bottom) are shown for first-stage ascent, upper-stage flight, and booster return. After stage separation, the upper and lower branches correspond to the upper stage and booster, respectively.

settings define the modeled mission and are not presented as independently established Falcon 9 operating limits.

Table 2 Prescribed/Estimated vehicle parameters and settings for the reference scenario.

Parameter	Value
First-stage inert mass [kg]	25,600
Upper-stage inert mass [kg]	4,000
First-stage propellant mass [kg]	395,700
Upper-stage propellant mass [kg]	92,670
Upper-stage minimum throttle factor	0.638
Upper-stage minimum-throttle specific-impulse factor	0.975
Coast duration after stage separation [s]	10
Maximum ascent dynamic pressure [kPa]	30
Maximum stage-separation dynamic pressure [kPa]	0.5
Maximum recovery dynamic pressure [kPa]	70
Maximum recovery heat flux [kW/m^2]	110
Maximum payload acceleration [g_0]	3.75
Fairing-separation altitude [km]	85
Ascent zero-angle drag coefficient	0.50
Booster-return drag coefficient	2.08

The Falcon 9 model calibrated using Starlink Group 15-27 is assessed against flight telemetry from three additional missions with different recovery strategy: SXM-11, SPHEREx and MRV. The parameters of each mission are given in table 3. For each case, the trajectory is reoptimized using the mission-specific payload mass, launch site, target orbit, and recovery mode, while the calibrated vehicle parameters are held fixed. This comparison assesses how well a single parameter set reproduces flight histories across different mission profiles. The missions span all three booster operating modes and include both direct insertion and insertion with an upper-stage restart.

Table 3 Falcon 9 missions used to assess the calibrated model. The listed orbital elements are public reference values for the deployment orbit.

Parameter	SXM-11	SPHEREx	MRV–MEP
Launch date (UTC)	29 Jun. 2026	11 Mar. 2025	21 Jul. 2026
Launch site	Cape Canaveral	Vandenberg	Cape Canaveral
Launch pad	SLC-40	SLC-4E	SLC-40
Payload mass [kg]	6950	758	4200
Orbit type	GTO	SSO	Supersynchronous GTO
Apogee altitude [km]	19,463[a]	660	44,765[b]
Perigee altitude [km]	204[a]	646	309[b]
Inclination [deg]	26.48[a]	97.96	25.63[b]
Booster recovery mode	ASDS	RTLS	Expendable
Upper-stage burns to deployment	2	2	2

[a] Prelaunch estimate of the SXM-11 injection orbit. [b] Early tracking estimate for the Mission Robotic Vehicle, approximately two days after launch. The MRV–MEP payload mass refers to the combined mass of the robotic vehicle and three Mission Extension Pods. GTO denotes geosynchronous transfer orbit; SSO denotes Sun-synchronous orbit.

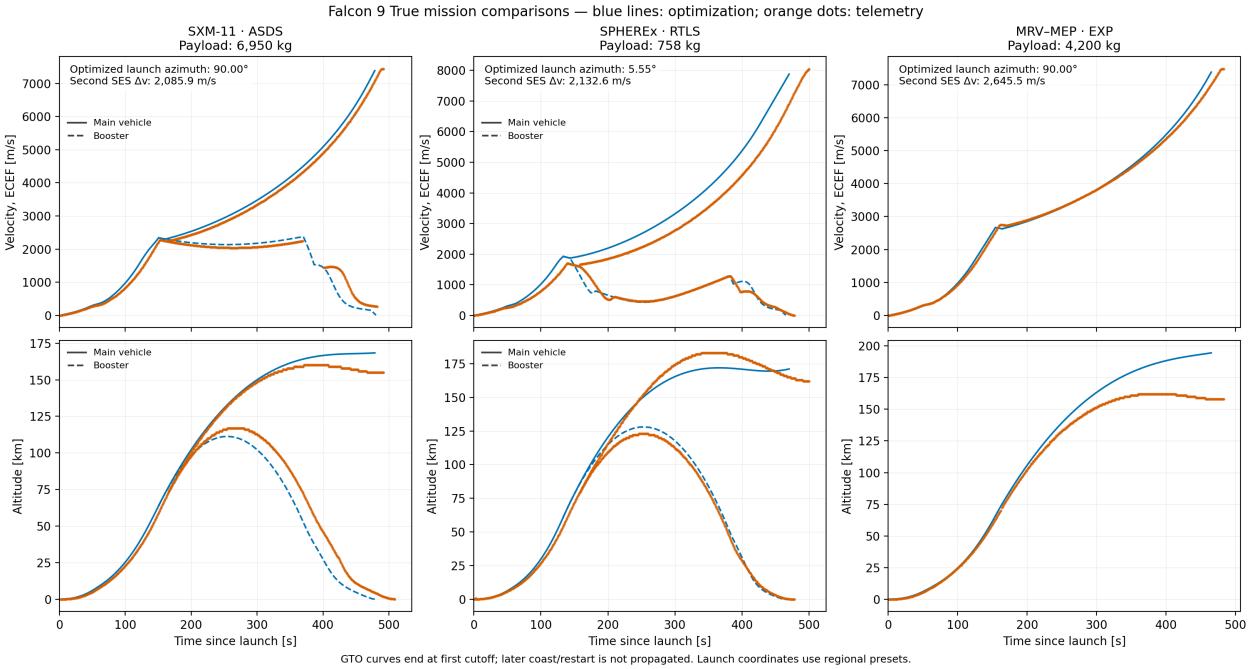


Fig. 3 Comparison of optimized trajectories (blue lines) with flight telemetry (orange markers) for SXM-11, SPHEREx, and MRV–MEP. Earth-fixed speed (top) and altitude (bottom) are shown versus time since liftoff. Solid blue lines denote the ascent trajectory, including upper-stage flight after separation. Dashed blue lines denote booster recovery for SXM-11 (downrange drone-ship landing) and SPHEREx (return to launch site). MRV–MEP was flown in expendable mode. For SXM-11 and MRV–MEP, the comparison ends at the first upper-stage cutoff.

Figure 3 compares the optimized altitude and Earth-fixed speed histories with flight telemetry for three Falcon 9 missions spanning downrange recovery, return to launch site, and expendable operation. For SXM-11, the optimized ascent closely follows the measured histories, although it ends at a higher altitude. The modeled booster reaches a slightly lower peak altitude and descends and lands earlier than indicated by telemetry. For SPHEREx, the optimized ascent generally runs ahead of the measured speed history and reaches a lower peak altitude. The modeled booster return follows the measured altitude closely after its peak, while differences in speed are most apparent around boostback and reentry. For MRV–MEP, the optimized speed follows telemetry closely through most of ascent, but the altitude histories diverge during upper-stage flight: the modeled trajectory reaches approximately 195 km at the first cutoff, compared with approximately 160 km in telemetry. This case shows that close agreement in speed can coexist with substantial differences in the altitude profile. These discrepancies may reflect differences in trajectory optimization objectives and guidance strategies, mission-specific operational constraints omitted from the formulation, and simplifications in the vehicle and environmental models.

VIII. Sensitivity Analysis

The sensitivity analysis evaluates how changes in vehicle parameters and operating limits affect the maximum deliverable payload. Two missions launched from Kennedy Space Center are considered: a circular low Earth orbit (LEO) at 250 km altitude and 30° inclination, and a Trans-Lunar Injection (TLI) at 28.6° inclination with a perigee altitude of 200 km and apogee of 384400 km. Each case is evaluated for expendable operation (EXP), downrange drone-ship recovery (ASDS), and return to launch site (RTLS).

For each mission and recovery mode, one parameter p_i is varied at a time, and the trajectory is reoptimized at both perturbed values. The local sensitivity of optimized payload mass is estimated by the central difference

$$S_i \approx \frac{m_{\mathrm{PL}}^{\star}(\mathbf{p}_0 + h_i \mathbf{e}_i) - m_{\mathrm{PL}}^{\star}(\mathbf{p}_0 - h_i \mathbf{e}_i)}{2h_i}, \tag{44}$$

where $\mathbf{p}_0$ is the baseline parameter vector, h_i is the perturbation magnitude, and $\mathbf{e}_i$ selects the perturbed parameter. Table 4 reports the resulting payload changes per unit increase in each parameter. Specific impulse and thrust are expressed per 1% increase because the study scales each stage's propulsion parameters by a common factor. In particular, the first-stage specific-impulse perturbation scales both the sea-level and vacuum values.

The recovery strategy changes the relative value of additional propellant in the two stages. From EXP to RTLS, the payload gain per kilogram of added first-stage propellant decreases, while the gain per kilogram of added upper-stage propellant increases. In LEO, the ratio of upper-stage to first-stage propellant sensitivity rises from approximately 1.1 in EXP to 5.8 in RTLS. The high-apogee case shows the same trend. These sensitivities describe independent changes in stage propellant mass; they do not include the additional tank mass needed to carry more propellant or imply the same

Table 4 Baseline optimized payload and selected local payload sensitivities. Sensitivity units give the change in payload mass per unit increase in the listed parameter. The stage-partition sensitivity is expressed per percentage point of the model's partition fraction.

		LEO			High-apogee transfer		
Quantity	Unit	EXP	ASDS	RTLS	EXP	ASDS	RTLS
Baseline payload	t	19.65	17.40	13.93	4.95	3.95	2.58
Recovery penalty relative to EXP	%	—	11.5	29.1	—	20.2	47.9
First-stage specific impulse	kg per 1%	308	253	178	116	90	60
Upper-stage specific impulse	kg per 1%	375	364	347	220	207	185
First-stage thrust	kg per 1%	81	64	50	25	19	15
Upper-stage thrust	kg per 1%	43	46	32	9	9	7
First-stage inert mass	kg/kg	−0.138	−0.285	−0.278	−0.054	−0.106	−0.101
Upper-stage inert mass	kg/kg	−1.000	−1.000	−1.000	−1.000	−1.000	−1.000
First-stage propellant mass	kg/kg	0.031	0.025	0.016	0.012	0.009	0.005
Upper-stage propellant mass	kg/kg	0.036	0.059	0.093	0.027	0.037	0.044
Stage-partition fraction	kg/percentage point	+140.6	−51.0	−257.1	+115.6	+36.3	−13.0
Launch altitude	kg/km	416	331	165	148	111	52
Ascent dynamic-pressure limit	kg/kPa	49.0	38.5	21.0	18.1	13.4	6.9
Recovery dynamic-pressure limit	kg/kPa	—	33.7	33.3	—	11.5	11.9

gain from reallocating propellant at fixed liftoff mass.

Among the propulsion parameters tested, upper-stage specific impulse gives the largest payload gain for a 1% increase in every case. For RTLS, the gain falls from 347 kg in LEO to 185 kg in the high-apogee case, but rises from 2.49% to 7.18% of the corresponding baseline payload. The recovery penalty shows a similar distinction: RTLS costs less payload in absolute terms in the high-apogee case (2.37 t versus 5.72 t in LEO), but consumes a larger fraction of expendable capacity (47.9% versus 29.1%).

In both missions, an additional kilogram of upper-stage inert mass displaces approximately one kilogram of payload. The payload penalty per kilogram of first-stage inert mass is about twice as large for the recovered cases as for EXP. The recovery dynamic-pressure limit also has a measurable effect, but the response need not be symmetric. For LEO with ASDS recovery, raising the modeled limit from 70 to 77 kPa adds 191 kg of payload, whereas lowering it to 63 kPa costs 280 kg. This difference illustrates why the central slope is useful for comparing sensitivities, while the individual perturbed solutions are more informative when evaluating a specific change to an operating limit.

The stage-partition sensitivity decreases from EXP to ASDS to RTLS in both missions. Increasing the partition fraction benefits EXP but reduces payload for RTLS. ASDS lies between these two modes and changes sign between missions: the sensitivity is −51.0 kg per percentage point in LEO and +36.3 kg per percentage point in the high-apogee transfer. The RTLS response is also strongly mission dependent, falling in magnitude from 257.1 to 13.0 kg per percentage point. Thus, a partition change that improves expendable performance can reduce payload when the same booster must return to the launch site. The partition should therefore be evaluated with the intended mission and recovery mode included in the optimization. The tabulated values are local slopes estimated using perturbations of ±0.1

percentage points.

The cross-mission results compare two modeled cases. Because their target perigees and inclinations differ, and the historical LEO run does not retain a complete vehicle and solver snapshot, the differences between missions cannot be attributed to injection energy alone.

A. Using Sensitivities to Screen Design Changes

A design change can affect several vehicle parameters at once. A first-order estimate of the combined payload effect can be obtained by multiplying each parameter change by its local sensitivity and summing the contributions. This estimate helps identify which concepts warrant a full trajectory reoptimization.

As an example, consider adding a tenth Merlin 1D engine to a first stage with nine identical engines. Assuming the same thrust and specific impulse per engine, the additional engine increases rated first-stage thrust by $1/9$, or 11.1%. The Merlin 1D dry mass of 470 kg is used as the engine-only increment in first-stage inert mass. The installed mass would also include any changes to the thrust structure, propellant feed system, and other supporting hardware.

Let S_T be the payload sensitivity in kilograms per 1% increase in rated first-stage thrust, and let S_m be the payload sensitivity in kilograms per kilogram of first-stage inert mass. The estimated net payload change is

$$\Delta m_{\mathrm{PL}} \approx S_T \frac{100}{9} + S_m \Delta m_{\mathrm{installed}}. \tag{45}$$

Table 5 evaluates this expression using $\Delta m_{\mathrm{installed}} = 470\,\mathrm{kg}$ as an engine-only estimate and holding the other vehicle parameters fixed.

Table 5 Estimated net payload change from an 11.1% increase in rated first-stage thrust and a $470\,\mathrm{kg}$ increase in first-stage inert mass.

Mission	EXP [kg]	ASDS [kg]	RTLS [kg]
LEO	+838	+574	+429
High-apogee transfer	+256	+167	+124

The estimated net effect is positive in all six cases. In LEO, the additional thrust contributes approximately 903 kg of payload in EXP and 560 kg in RTLS, while the engine dry mass offsets approximately 65 and 131 kg, respectively. The net RTLS gain is therefore about half the expendable gain: the additional thrust is less valuable, and carrying the additional engine is more costly in payload terms.

IX. Software Availability and Graphical User Interface

To facilitate the application and further development of the proposed framework, the launch-vehicle trajectory optimization software is made available through a public GitHub repository:

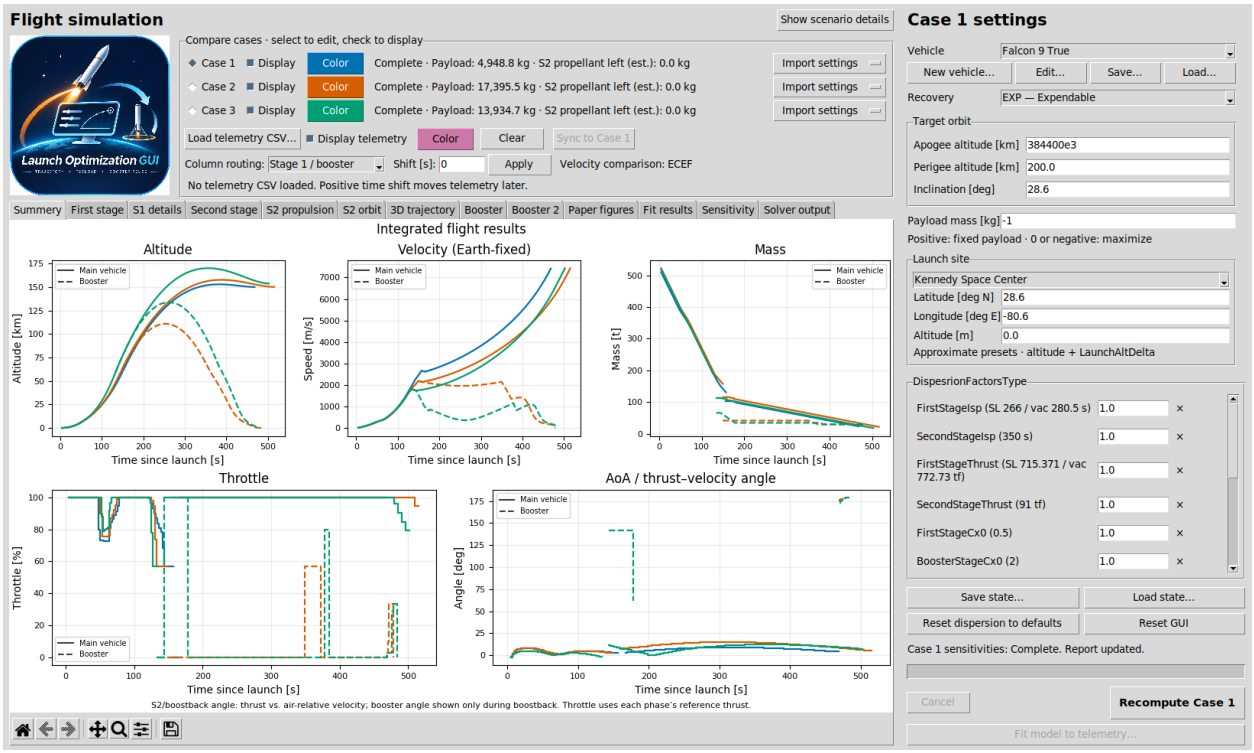


Fig. 4 Graphical user interface of the launch-vehicle trajectory optimization framework. The interface supports vehicle and mission configuration, selection of booster recovery strategies, and comparison of multiple optimization scenarios.

https://github.com/pukmark/LaunchVehicleOptimization

The software includes a graphical user interface (GUI) that allows users to configure launch-vehicle parameters, specify mission requirements, and execute trajectory optimizations without directly modifying the underlying optimization code. Users can define vehicle characteristics, including stage masses, propulsion parameters, and aerodynamic properties; prescribe target orbital conditions and launch-site parameters; and select among expendable, ASDS, and RTLS recovery strategies.

The GUI provides access to the optimization results, enabling users to examine the resulting trajectories and evaluate vehicle performance under different mission and recovery configurations. By modifying individual vehicle parameters and repeating the optimization, users can also investigate design tradeoffs and perform sensitivity studies similar to those presented in Section VIII.

The repository is intended to support reproducible research and provide a starting point for further investigations in launch-vehicle trajectory optimization and preliminary vehicle design. Researchers are encouraged to download the software, apply it to alternative vehicle configurations and mission scenarios, and extend the framework for their own research applications.

X. Conclusions

This work presented a trajectory optimization framework for assessing the payload performance of two-stage launch vehicles with recoverable boosters. The framework jointly optimizes ascent, orbital injection, and booster recovery, allowing expendable operation, downrange recovery, and return to the launch site to be compared for the same mission. A Falcon 9 model calibrated using Starlink Group 15-27 telemetry reproduced the main altitude and Earth-fixed speed trends for three additional missions spanning all three booster operating modes. The comparisons support the use of the framework for mission-level trade studies, while leaving propellant consumption, recovery load margins, and the final injection maneuver on transfer-orbit missions unverified by telemetry.

For the modeled 250 km low Earth orbit mission, downrange recovery and return to the launch site reduced the optimized payload by 11.5% and 29.1%, respectively, relative to expendable operation. Recovery also changed the value of vehicle improvements: as recovery requirements became more demanding, the marginal payload benefit of adding first-stage propellant fell, while that of adding upper-stage propellant rose. Upper-stage specific impulse produced the largest payload gain among the propulsion parameters tested for equal percentage changes. The preferred stage partition also depended on the recovery strategy: a partition change that improved expendable performance reduced payload in the return-to-launch-site cases.

The additional-engine example illustrated how local sensitivities can screen design changes that affect thrust and inert mass simultaneously. The first-order estimate indicated a net payload benefit in all modeled cases, with a smaller benefit when the booster was recovered. A full reoptimization with the installed engine mass and revised operating assumptions is needed to confirm that estimate. Together, these results show why mission and recovery requirements should be included when ranking vehicle design changes early in development.

A. Appendix I

The procedure to transform the position and velocity from the local CS used in the boost phase (LLAA) to the CS used in the second phase which is Earth Center Inertial (ECI) given launch azimuth is presented. The main differences between the coordinate systems are the position of the origin and the fact that ECI is inertial and LLAA is not. The main idea is to build a North-East-Up triad at the launch site, rotate the local state into ECI using the given azimuth, and then add the site's ECI position and spin velocity—yielding the full inertial state to start the second-stage/ECI dynamics.

- Calculate the ECI position and velocity of the launch site based on the launch site Latitude (Lat_{LS}), Longitude

(Lon_{LS}) and Altitude (Alt_{LS}):

$$\vec{P}_{LS} = \begin{bmatrix} (R_e + Alt_{LS})\cos(Lat_{LS})\cos(Lon_{LS}) \\ (R_e + Alt_{LS})\cos(Lat_{LS})\sin(Lon_{LS}) \\ (R_e + Alt_{LS})sin(Lat_{LS}) \end{bmatrix}$$
$$\vec{V}_{LS} = \omega_e \begin{bmatrix} (R_e + Alt_{LS})\cos(Lat_{LS})\cos(Lon_{LS}) \\ -(R_e + Alt_{LS})\cos(Lat_{LS})\sin(Lon_{LS}) \\ 0 \end{bmatrix} \tag{46}$$

- build the North-East-Up vectors in ECI frame:

$$\begin{aligned} \hat{z}_{up} &= [cos(Lat_{LS})cos(Lon_{LS}) \quad cos(Lat_{LS})sin(Lon_{LS}) \quad sin(Lat_{LS})]^T \\ \hat{x}_{north} &= [-sin(Lat_{LS})cos(Lon_{LS}) \quad -sin(Lat_{LS})sin(Lon_{LS}) \quad cos(Lat_{LS})]^T \\ \hat{y}_{east} &= [-sin(Lon_{LS}) \quad cos(Lon_{LS}) \quad 0]^T \end{aligned} \tag{47}$$

- Rotate the north vector by angle Az to get the x vector pointed to the flight azimuth. y vector completes a right-hand CS with the Up vector :

$$\begin{aligned} \hat{x}_{Az} &= cos(Az)\hat{x}_{north} + sin(Az)\hat{y}_{east} \\ \hat{y}_{Az} &= \hat{z}_{up} \times \hat{x}_{Az}\hat{z}_{Az} \qquad\qquad = \hat{z}_{up} \end{aligned} \tag{48}$$

- The position and velocity in ECI frame as function of the position (($\vec{P}_{LLAA}$) and velocity (($\vec{V}_{LLAA}$) in LLAA frame :

$$\begin{aligned} \vec{P}_{ECI} &= P_{LS} + [\hat{x}_{Az}, \hat{y}_{Az}, \hat{z}_{Az}]\vec{P}_{LLAA} \\ \hat{V}_{ECI} &= V_{LS} + [\hat{x}_{Az}, \hat{y}_{Az}, \hat{z}_{Az}]\vec{V}_{LLAA} \end{aligned} \tag{49}$$

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